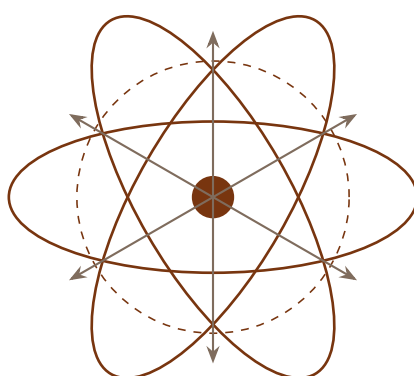


SPhOH

SAUDI PHYSICS OLYMPIAD HUB



Vectors

Olympiad Problem Set

Olympiad Training • Math

SPhOH Team

FIRST EDITION • 2026

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About This Problem Set

This problem set is part of the training resources of the **Saudi Physics Olympiad Hub (SPhOH)**.

It is designed primarily for students preparing for the Saudi Physics Olympiad and related advanced physics competitions. The purpose of the handout is to provide a focused and substantial collection of problems for deliberate problem-solving practice.

The handout is **problem-centered**. Explanations are intentionally brief and are included only where they help establish the ideas or techniques needed for the problems.

How to Use the Problem Set

Students are encouraged to attempt each problem independently before looking at its hint. A hint should be treated as a way to overcome a specific obstacle, not as a substitute for solving the problem.

For difficult problems, several progressive hints may be provided. Try to use the least amount of assistance necessary before returning to the problem.

Difficulty Levels

The normal problems are divided into three training levels:

- **Level I** — Easy and direct substitution problems.
- **Level II** — Problems requiring combinations of ideas and more substantial reasoning.
- **Level III** — National Olympiad level problems.

Find more at: <https://devfhd.github.io/SPhOH/>.

CHAPTER 1

Vectors

Topic Note.

Vectors represent quantities possessing both magnitude and direction. Master the decomposition into Cartesian components, operations in unit vector notation ($\hat{i}, \hat{j}, \hat{k}$), dot products (scalar projections and angles), and cross products (orthogonal vectors and geometric areas).

1.1 Vector Components

PROBLEM 1

LEVEL I

Vector Component Magnitudes

Find $\tan(\theta)$ (measured from the positive horizontal axis) for a vector that has identical magnitudes in its x and y components.

PROBLEM 2

LEVEL I

Angle from Vertical Axis

Find $\sin(\theta)$ (measured from the positive y -axis) for a vector whose x -component is 3 times larger than its y -component.

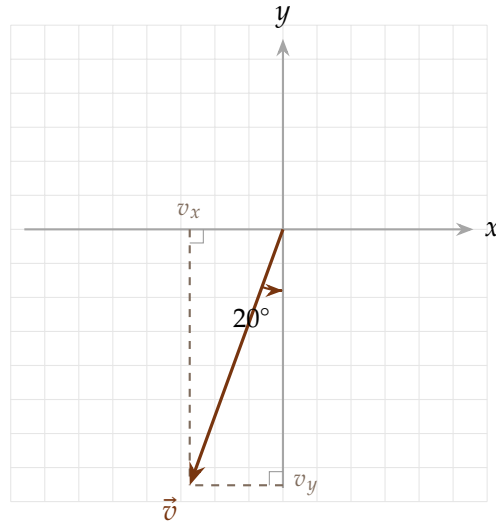
PROBLEM 3

LEVEL I

Polar Components and Sketch

What are the components of a vector with a magnitude of 10 units and a direction of 20°

clockwise from the negative y -axis (-20° from the negative y -axis)? Sketch the vector on a 2D coordinate plane.



PROBLEM 4

LEVEL I

Position Vector Orientation

A position vector has components $d_x = -3$ m and $d_y = 4$ m.

- (a) Find its magnitude.
- (b) Find its direction relative to the positive x -axis.
- (c) Find its direction relative to the negative y -axis.

PROBLEM 5

LEVEL I

Force Vector Decomposition

A force vector $\vec{F}$ has a magnitude of 25 N and points at an angle of 120° relative to the positive x -axis. Determine its scalar components F_x and F_y .

1.2 Unit Vectors

PROBLEM 6

LEVEL I

Unit Vector Notation of Force

A force vector with a magnitude of 10 N is directed at an angle $\theta = 45^\circ$ from the positive x -axis. Express this force vector in terms of unit vectors $\hat{i}$, $\hat{j}$, and $\hat{k}$.

PROBLEM 7

LEVEL II

3D Velocity and Direction Angles

A velocity vector is given by $\vec{v} = (-5\hat{i} - 3\hat{j} + 2\hat{k})$ m/s. Find:

- (a) The magnitude of the velocity vector.
- (b) The direction angles between the vector and all three coordinate axes (x , y , and z).
- (c) Sketch the vector along with its component projections on a 3D xyz -coordinate system.

PROBLEM 8

LEVEL II

Acceleration Vector from Component Ratios

Express an acceleration vector with a magnitude of 10 m/s^2 that satisfies the component relationship $a_y = 2a_z = 3.5a_x$. Furthermore, calculate the direction angles between the acceleration vector and all three coordinate axes.

PROBLEM 9

LEVEL III

Jerk Vector Representation

Jerk ($\vec{j}$) is defined as the time rate of change of acceleration. Express the jerk vector in unit vector notation under the following conditions:

- (a) Motion with constant acceleration.
- (b) The components satisfy $2j_y = 0.235j_z$, with $j_x = 0$.
- (c) The magnitude of the jerk vector is 10 m/s^3 , $j_y = 2 \text{ m/s}^3$, and $\sin(\theta) = 3/5$, where θ is the angle between the vector and the xz -plane.

1.3 Vector Addition and Subtraction

PROBLEM 10

LEVEL I

Displacement Bounds

A person walks twice, with displacement magnitudes of 8 m and 3 m, respectively.

- (a) What is the maximum possible magnitude of the net displacement?
- (b) What is the minimum possible magnitude of the net displacement?

PROBLEM 11

LEVEL I

Resultant of Two Vectors

Consider two displacement vectors $\vec{d}_1 = 3\hat{i} + \sqrt{55}\hat{j}$ and $\vec{d}_2 = -3\hat{i}$.

- (a) Sketch the vectors on the xy -plane.
- (b) Find the magnitude of the resultant vector $\vec{R} = \vec{d}_1 + \vec{d}_2$.
- (c) Determine the direction angle of the resultant vector measured from the negative x -axis.

PROBLEM 12

LEVEL II

Vector Subtraction in 3D

Vector $\vec{C}$ is the resultant vector formed by adding vector $\vec{A}$ and vector $\vec{B}$ ($\vec{C} = \vec{A} + \vec{B}$). Given $\vec{C} = 5\hat{i} + 3\hat{k}$ and $\vec{A} = 2\hat{j} + 4\hat{k}$:

- (a) Find vector $\vec{B}$ in unit vector notation.

- (b) Calculate the magnitude and direction of vector $\vec{B}$.

PROBLEM 13

LEVEL II

Law of Cosines and Pythagorean Theorem

Consider two vectors $\vec{A}$ and $\vec{B}$ with an angle θ between them.

- (a) Show that the magnitude of the resultant vector $\vec{C} = \vec{A} + \vec{B}$ can be written in the following form: $C^2 = A^2 + B^2 - 2AB \cos \theta$.
- (b) Using this resultant formula, prove the Pythagorean theorem.

PROBLEM 14

LEVEL II

Angle and Resultant Variation

Two vectors have magnitudes $A = 5 \text{ m/s}$ and $B = 10 \text{ m/s}$.

- (a) Find the angle θ between vectors $\vec{A}$ and $\vec{B}$ if the magnitude of their resultant $\vec{C}$ is 11 m/s .
- (b) Find the resultant magnitude if $\theta = 90^\circ$.
- (c) Find the resultant magnitude if $\theta = -180^\circ$.
- (d) Find the resultant magnitude if $\theta = -45^\circ$.

PROBLEM 15

LEVEL III

Algebraic Vector System and Projections

Given the vectors:

$$\begin{aligned}\vec{A} &= x\hat{i} + (4y - 2)\hat{j} - 5\hat{k} \\ \vec{B} &= (-2x)\hat{i} + 2\hat{j} + (1.5z + 4)\hat{k} \\ \vec{C} &= \vec{A} - \vec{B} = (9 + x)\hat{i} + (7 - 2y)\hat{j} - 8\hat{k}\end{aligned}$$

- (a) Solve for the unknown variables x , y , and z .
- (b) Draw vector $\vec{A}$ on the xy -plane.
- (c) Draw vector $\vec{B}$ on the zy -plane.

- (d) Draw the y -component of $\vec{C}$ on the xz -plane.

PROBLEM 16**LEVEL III****Three-Variable Vector Subtraction System**

Given the vectors:

$$\vec{A} = 2a\hat{i} + 3b\hat{j} - c\hat{k}$$

$$\vec{B} = a\hat{i} - b\hat{j} + 2c\hat{k}$$

$$\vec{C} = 3\hat{i} + 2\hat{j} - 4\hat{k}$$

$$\vec{D} = \vec{A} - \vec{B} - \vec{C} = 5\hat{i} + 6\hat{j} + 3\hat{k}$$

- (a) Determine the scalar constants a , b , and c .
- (b) Express vectors $\vec{A}$ and $\vec{B}$ explicitly in unit vector notation.
- (c) Calculate the magnitude of vector $\vec{D}$.
- (d) Draw vector $\vec{C}$ on the xz -plane.

1.4 Dot Product

Topic Note: Scalar (Dot) Product

The dot product of two vectors $\vec{A}$ and $\vec{B}$ is a scalar quantity defined as:

$$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}| \cos(\theta) = A_x B_x + A_y B_y + A_z B_z$$

Key applications include computing work $W = \vec{F} \cdot \vec{d}$, testing orthogonality ($\vec{A} \cdot \vec{B} = 0$), and finding projections.

PROBLEM 17

LEVEL I

Angle Between Two Vectors

Given two vectors $\vec{A} = 3\hat{i} + 4\hat{j}$ and $\vec{B} = -2\hat{i} + 5\hat{j}$, calculate:

- (a) The dot product $\vec{A} \cdot \vec{B}$.
- (b) The angle θ between vectors $\vec{A}$ and $\vec{B}$.

PROBLEM 18

LEVEL I

Self-Dot Product Properties

What is the scalar result of forming the dot product of a general vector $\vec{V} = V_x\hat{i} + V_y\hat{j} + V_z\hat{k}$ with itself ($\vec{V} \cdot \vec{V}$)? Express your answer in terms of the magnitude $V = |\vec{V}|$.

PROBLEM 19

LEVEL II

Orthogonality Condition

Determine the value of the constant c such that the vectors $\vec{v}_1 = 2\hat{i} + c\hat{j} - 3\hat{k}$ and $\vec{v}_2 = c\hat{i} + 4\hat{j} + 2\hat{k}$ are perpendicular to each other.

PROBLEM 20

LEVEL II

Work Done by a Constant Force

A constant force $\vec{F} = (6\hat{i} - 2\hat{j} + 4\hat{k})$ N acts on an object, displacing it by $\vec{d} = (3\hat{i} + 5\hat{j} + 2\hat{k})$ m.

- (a) Calculate the work done $W = \vec{F} \cdot \vec{d}$ by the force.
- (b) Determine the angle θ between the force vector and the displacement vector.

PROBLEM 21

LEVEL III

Vector Projection

Given vectors $\vec{p} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{q} = 3\hat{i} + 4\hat{k}$:

- (a) Find the scalar projection of $\vec{p}$ onto $\vec{q}$.
- (b) Find the vector projection $\vec{q}$ onto $\vec{p}$.
- (c) compare the results of a and b.

1.5 Cross Product

Topic Note: Vector (Cross) Product

The cross product of two vectors $\vec{A}$ and $\vec{B}$ yields a vector perpendicular to both, defined by the determinant:

$$\vec{A} \times \vec{B} = (A_y B_z - A_z B_y)\hat{i} + (A_z B_x - A_x B_z)\hat{j} + (A_x B_y - A_y B_x)\hat{k}$$

Its magnitude is $|\vec{A} \times \vec{B}| = |\vec{A}||\vec{B}|\sin(\theta)$, which equals the area of the parallelogram formed by $\vec{A}$ and $\vec{B}$.

PROBLEM 22

LEVEL I

Cross Product Calculation

Compute the cross product $\vec{A} \times \vec{B}$ for the vectors $\vec{A} = 2\hat{i} - \hat{j} + 3\hat{k}$ and $\vec{B} = 4\hat{i} + 2\hat{j} - \hat{k}$.

PROBLEM 23

LEVEL I

Self-Cross Product Properties

What is the vector result of taking the cross product of any arbitrary vector $\vec{V}$ with itself ($\vec{V} \times \vec{V}$)? Explain using both the magnitude equation and component definitions.

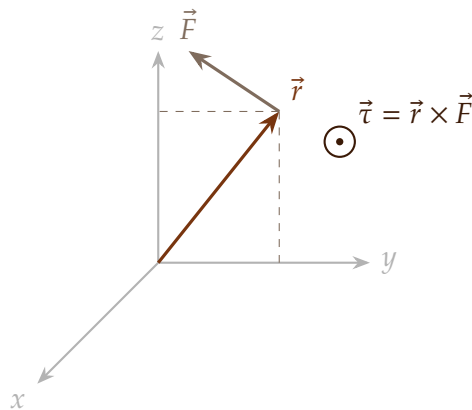
PROBLEM 24

LEVEL II

Torque

Torque is a vector defined to be the cross product of moment arm and force, which can be written as: $\vec{\tau} = \vec{r} \times \vec{F}$. Consider that $\vec{r} = 4\hat{i} + 3\hat{j} + 5.5\hat{k} \text{ m}$, and $\vec{F} = -6\hat{i} + 12\hat{j} \text{ N}$. Find the torque in:

- (a) unit vectors.
- (b) direction and magnitude.

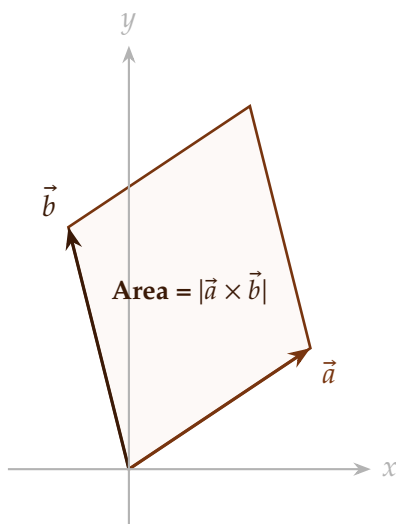


PROBLEM 25

LEVEL II

Parallelogram Area

Find the area of the parallelogram defined by the vectors $\vec{a} = 3\hat{i} + 2\hat{j}$ and $\vec{b} = -\hat{i} + 4\hat{j}$.

**PROBLEM 26****LEVEL III****Unit Normal Vector to a Plane**

Find a unit vector perpendicular to both $\vec{u} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{v} = \hat{i} + 3\hat{j} + \hat{k}$.

PROBLEM 27**LEVEL III****Three-Variable Cross Product Relation**

Consider three non-zero vectors in 3D Cartesian space satisfying the vector product equation $\vec{A} \times \vec{B} = \vec{C}$, where:

$$\vec{A} = 2\hat{i} + y\hat{j} + 0\hat{k}$$

$$\vec{B} = 0\hat{i} + 3\hat{j} + z\hat{k}$$

$$\vec{C} = 12\hat{i} - 4\hat{j} + 6\hat{k}$$

- (a) Determine the unknown scalar constants y and z .
- (b) Verify that $\vec{C}$ is orthogonal to both $\vec{A}$ and $\vec{B}$ using dot products.
- (c) Calculate the area of the triangle formed by vectors $\vec{A}$ and $\vec{B}$.
- (d) Find the sine of the angle θ between vectors $\vec{A}$ and $\vec{B}$.

CHAPTER 2

Hints

The hints below correspond to the problems in the handout. They are provided separately so that students can work through the problem collection without immediately seeing the associated guidance.

Hint 1 | Problem 1 – Vector Component Magnitudes

Recall that $\tan(\theta) = v_y/v_x$. Since $|v_x| = |v_y|$, consider both equal signs and opposite signs ($v_y = \pm v_x$).

Hint 2 | Problem 2 – Angle from Vertical Axis

If θ is measured from the positive y -axis, then $\tan(\theta) = |v_x/v_y| = 3$. Construct a right triangle or use $\sin(\theta) = \frac{\tan(\theta)}{\sqrt{1+\tan^2(\theta)}}$.

Hint 3 | Problem 3 – Polar Components and Sketch

The negative y -axis corresponds to an angle of 270° . Moving 20° clockwise places the vector in the third quadrant at an angle of 250° from the positive x -axis.

Hint 4 | Problem 4 – Position Vector Orientation

The magnitude is $|\vec{d}| = \sqrt{d_x^2 + d_y^2}$. For the direction, note that $d_x < 0$ and $d_y > 0$, placing the vector in Quadrant II.

Hint 5 | Problem 5 – Force Vector Decomposition

Use $F_x = F \cos(\theta)$ and $F_y = F \sin(\theta)$ with $\theta = 120^\circ$.

Hint 6 | Problem 6 – Unit Vector Notation of Force

Because the force lies in the xy -plane, $F_z = 0$. Use $F_x = F \cos(45^\circ)$ and $F_y = F \sin(45^\circ)$.

Hint 7 | Problem 7 – 3D Velocity and Direction Angles

Direction cosines satisfy $\cos(\alpha) = v_x/v$, $\cos(\beta) = v_y/v$, and $\cos(\gamma) = v_z/v$.

Hint 8 | Problem 8 – Acceleration Vector from Component Ratios

Express a_y and a_z in terms of a_x : $a_y = 3.5a_x$ and $a_z = 1.75a_x$. Substitute these into $a^2 = a_x^2 + a_y^2 + a_z^2 = 10^2$ to solve for a_x .

Hint 9 | Problem 9 – Jerk Vector Representation

For part (a), if acceleration is constant, its derivative with respect to time is zero. For part (c), recall that the angle with the xz -plane satisfies $\sin(\theta) = |j_y|/j$.

Hint 10 | Problem 10 – Displacement Bounds

Use the triangle inequality: $|d_1 - d_2| \leq R \leq d_1 + d_2$.

Hint 11 | Problem 11 – Resultant of Two Vectors

Add the components: $\vec{R} = (3 - 3)\hat{i} + \sqrt{55}\hat{j} = \sqrt{55}\hat{j}$. The vector lies purely along the positive y -axis.

Hint 12 | Problem 12 – Vector Subtraction in 3D

Rearrange to $\vec{B} = \vec{C} - \vec{A} = (5 - 0)\hat{i} + (0 - 2)\hat{j} + (3 - 4)\hat{k}$.

Hint 13 | Problem 13 – Law of Cosines and Pythagorean Theorem

Expand $C^2 = (\vec{A} + \vec{B}) \cdot (\vec{A} + \vec{B}) = A^2 + B^2 + 2AB \cos(\theta)$. For part (b), set $\theta = 90^\circ$.

Hint 14 | Problem 14 – Angle and Resultant Variation

Use $C = \sqrt{A^2 + B^2 + 2AB \cos(\theta)}$ and solve for $\cos(\theta)$ in part (a).

Hint 15 | Problem 15 – Algebraic Vector System and Projections

Equate corresponding components of $\vec{A} - \vec{B}$ with $\vec{C}$. For the x -component: $x - (-2x) = 9 + x \implies 2x = 9$.

Hint 16 | Problem 16 – Three-Variable Vector Subtraction System

Equate corresponding components of $\vec{A} - \vec{B} - \vec{C} = \vec{D}$. For the x -component: $(2a - a) - 3 = 5 \implies a = 8$.

Hint 17 | Problem 17 – Angle Between Two Vectors

Compute $\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y$, then use $\cos(\theta) = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}||\vec{B}|}$.

Hint 18 | Problem 18 – Self-Dot Product Properties

Use the geometric definition $\vec{V} \cdot \vec{V} = |\vec{V}||\vec{V}| \cos(\theta)$ with $\theta = 0^\circ$, or sum the component products $V_x^2 + V_y^2 + V_z^2$.

Hint 19 | Problem 19 – Orthogonality Condition

Two non-zero vectors are perpendicular if and only if their dot product equals zero ($\vec{v}_1 \cdot \vec{v}_2 = 0$).

Hint 20 | Problem 20 – Work Done by a Constant Force

Compute $W = F_x d_x + F_y d_y + F_z d_z$ and use $W = Fd \cos(\theta)$ to find θ .

Hint 21 | Problem 21 – Vector Projection

The scalar projection is $\frac{\vec{p} \cdot \vec{q}}{|\vec{q}|}$ and the vector projection is $\left(\frac{\vec{p} \cdot \vec{q}}{|\vec{q}|^2} \right) \vec{q}$.

Hint 22 | Problem 22 – Cross Product Calculation

Expand the 3×3 determinant along the first row.

Hint 23 | Problem 23 – Self-Cross Product Properties

Recall that the angle between any vector and itself is $\theta = 0^\circ$. Evaluate $|\vec{V} \times \vec{V}| = |\vec{V}||\vec{V}| \sin(0^\circ)$.

Hint 24 | Problem 24 – Torque

Recall that:

$$\vec{A} \times \vec{B} = (A_y B_z - A_z B_y)\hat{i} + (A_z B_x - A_x B_z)\hat{j} + (A_x B_y - A_y B_x)\hat{k}$$

Hint 25 | Problem 25 – Parallelogram Area

The area of the parallelogram is given by the magnitude of the cross product $|\vec{a} \times \vec{b}|$.

Hint 26 | Problem 26 – Unit Normal Vector to a Plane

Compute $\vec{n} = \vec{u} \times \vec{v}$ and then normalize it: $\hat{n} = \frac{\vec{n}}{|\vec{n}|}$.

Hint 27 | Problem 27 – Three-Variable Cross Product Relation

Compute $\vec{A} \times \vec{B} = (yz)\hat{i} - (2z)\hat{j} + 6\hat{k}$ and match components with $\vec{C}$. The triangle area is $\frac{1}{2}|\vec{C}|$.